Case G — Mr. Arthur Chen | 69M | Idiopathic Pulmonary Fibrosis (IPF)

Retired stonemason. Dry non-productive cough, progressive breathlessness now limiting dressing. Digital clubbing. Fine velcro crackles both lung bases. Restrictive spirometry, reduced lung volumes. Task: assess functional capacity; provide breathlessness management and energy conservation.

Cardiorespiratory Physiotherapy

■ SUBJECTIVE PROBLEM LIST

- Physical Impairments
- Reduced lung compliance (stiff fibrotic lungs)
- Impaired gas exchange — reduced DLCO
- Reduced respiratory muscle endurance
- Progressive deconditioning and reduced aerobic capacity
- Breathlessness limiting dressing and personal care
- Reduced walking tolerance and community mobility
- Difficulty with stairs and inclines
- Reduced participation in leisure / social activities
- Anxiety related to breathlessness (breathlessness-fear cycle)

■■ RED FLAGS

- ■ SpO2 <88% at rest or during minimal activity
- Significant hypoxaemia. Impaired
- alveolar-capillary diffusion in IPF drops rapidly with exertion, risking
- arrhythmia and organ injury.
- Action: Stop assessment immediately.
- Position upright, apply supplemental O2 if available. Refer urgently to
- medical team. Do not exercise without medical clearance and O2
- ■ Acute exacerbation of IPF — sudden severe worsening of dyspnoea over days to weeks without identifiable cause

■ PATHOPHYSIOLOGY (KEY POINTS)

- IPF is a chronic, progressive interstitial lung disease characterised by abnormal wound-healing responses in the alveolar epithelium.
- Repeated micro-injuries to type II pneumocytes trigger aberrant fibroblast and myofibroblast activation, leading to excessive collagen deposition and parenchymal remodelling.
- This fibrosis thickens and stiffens the alveolar walls, dramatically reducing lung compliance — the lungs require far greater muscular effort to inflate for the same volume of air.
- The thickened alveolar-capillary membrane increases the diffusion distance for oxygen, impairing gas exchange and causing hypoxaemia that worsens substantially on exertion.
- The restrictive spirometric pattern (reduced TLC, FVC, and DLCO with preserved or elevated FEV1/FVC ratio) directly reflects this loss of functional parenchyma.
- Over time, hypoxic pulmonary vasoconstriction drives pulmonary hypertension, which can lead to right ventricular failure (cor pulmonale).

■ DIFFERENTIAL DIAGNOSES

1. Typically more steroid-responsive and carries a better prognosis.

■ OBJECTIVE ASSESSMENTS & HOW TO PERFORM

Six-Minute Walk Test (6MWT)

What it measures: Submaximal functional exercise capacity; identifies exertional desaturation; provides a prognostic marker and baseline for pulmonary rehab outcome monitoring in IPF.

Auscultation findings expected:

Fine, bilateral, late-inspiratory basal velcro crackles — the hallmark of IPF (stiff fibrotic alveoli snapping open). No wheeze. Globally reduced breath sounds reflecting restricted volumes. Post-exercise crackles may increase briefly as previously underventilated alveoli recruit.

Outcome measure:

6MWT distance in metres — repeat at 6 and 12 weeks. Also use K-BILD (King's Brief Interstitial Lung Disease questionnaire) for quality of

Thoracic Expansion / Chest Excursion Measurement

What it measures: Quantifies reduction restrictive mechanics and provides a baseline for monitoring the impact of breathing exercises. How to perform:

■ TREATMENT PLAN

- Action: Flag to referring respiratory
- and advance care planning rather than functional improvement.
- Airway Clearance
- Not indicated as a primary technique in IPF.
- Mr. Chen has a dry, non-productive cough — IPF does not produce excess
- mucus, so ACBT or PEP would not be appropriate and may exacerbate cough
- distress. If cough is particularly distressing, refer to respiratory
- physician for pharmacological cough suppression (e.g., low-dose

• morphine, thalidomide in palliative stage). The physiotherapy focus
• Breathing Exercise: Pursed-Lip Breathing (PLB) + Paced Breathing
• Breathe in through the nose for
• 2 counts. Breathe out slowly through pursed lips (as if cooling hot